ResumeIQ AI

Intelligent Resume Analyzer

GROUP –5

Problem Statement

Many students submit resumes without knowing: Is it ATS-friendly? Do skills match the target role? What improvements are needed?

Current Issues:

- *Manual review is slow and time-consuming*
- *Human errors occur frequently*
- *No role-based analysis available*
- *No proper feedback mechanism*

There is a critical need for an automated intelligent resume analysis system.

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Project Objectives

SMART ANALYSIS

Analyze uploaded resumes automatically with role-based skill matching to ensure candidates meet job requirements.

ATS SCORING

Calculate comprehensive ATS scores and detect missing technical skills for target roles.

SMART FEEDBACK

Generate actionable improvement suggestions and downloadable PDF reports for candidates.

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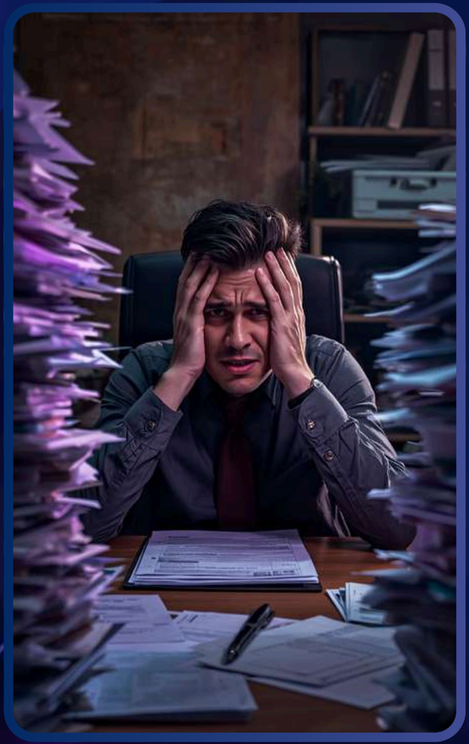
Existing vs Proposed System



EXISTING SYSTEM

- *Manual Review*
- *Slow Process*
- *Human Errors*
- *No ATS Check*
- *No Feedback*
- *No Report*

PROPOSED SYSTEM

- *Automated Analysis*
- *Instant Results*
- *Smart Scoring*
- *ATS Score*
- *Suggestions*
- *PDF Report*

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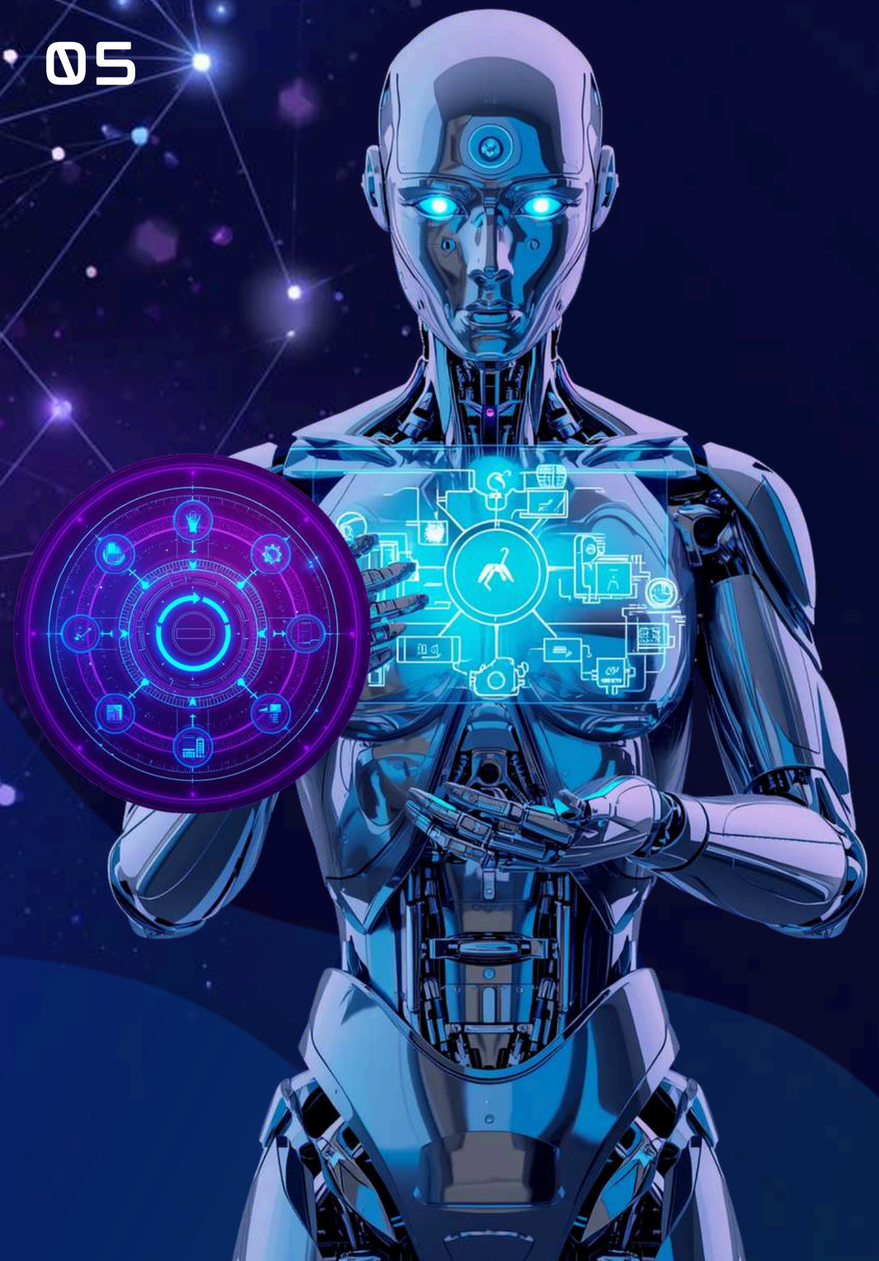

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System Architecture

User Upload Resume → Resume Parser → Skill Extraction → Role-Based Scoring → ATS Score Calculation → Feedback Generator → Result Dashboard → Dynamic PDF Report

This modular architecture ensures seamless data flow from initial resume upload through intelligent processing, delivering comprehensive analysis and actionable insights to users.

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Technology Stack

FRONTEND

HTML5 for structure, CSS3 for styling, and Jinja2 templating engine for dynamic content rendering.

BACKEND

Python with FastAPI framework for high-performance APIs, NLP parsing for text extraction.

TOOLS & LIBRARIES

ReportLab for PDF generation, VS Code IDE, Swagger UI for API documentation.

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Major Modules

UPLOAD & PARSING

Handles resume file uploads in PDF/DOCX formats. Extracts text content, contact details, and structures data for processing.

SCORING & FEEDBACK

Performs role-based skill matching, calculates ATS score, detects missing skills, and generates improvement suggestions.

DASHBOARD & REPORTS

Displays visual score breakdowns, progress bars, strength classification, and generates downloadable PDF reports.

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Key Features

ROLE-BASED ANALYSIS

Intelligent detection of missing skills based on target job roles. Compares your resume against role-specific requirements for precise gap analysis.

SMART ATS SCORE

Advanced weighted scoring system that evaluates resume compatibility with Applicant Tracking Systems for better visibility.

DYNAMIC PDF REPORTS

Resume strength classification with downloadable professional PDF reports containing detailed analysis and improvement suggestions.

Project Screenshots

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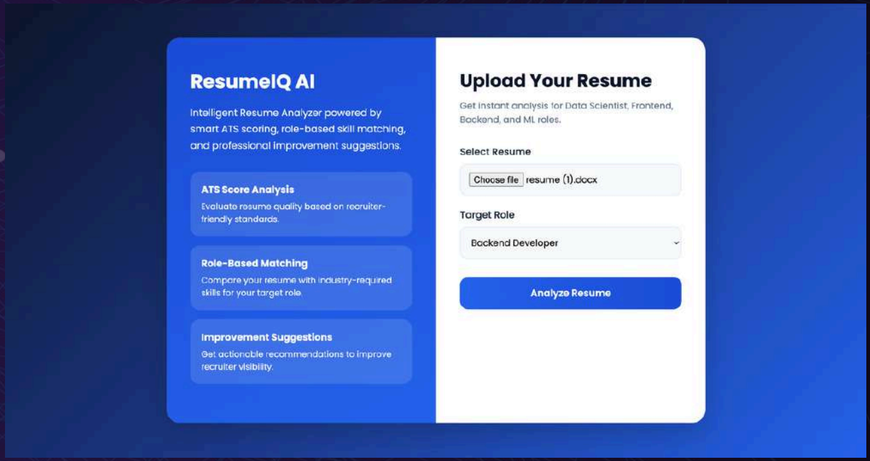
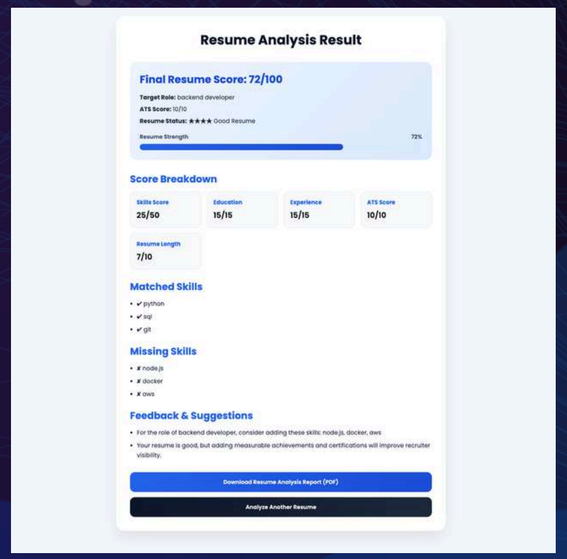
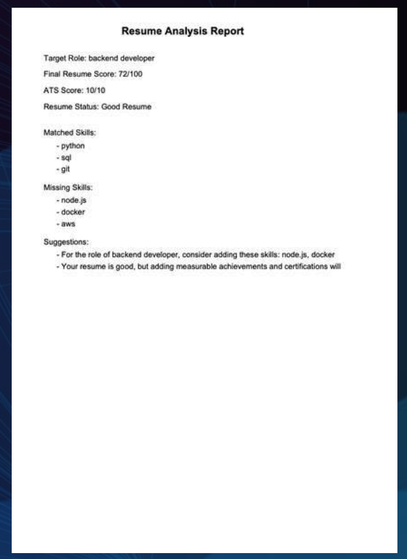
Our system features an intuitive user interface designed for seamless resume analysis. The Upload Resume Page provides a clean drag-and-drop interface for easy file submission. The Result Dashboard displays comprehensive ATS scores, skill breakdowns, and visual progress indicators. The PDF Report Download Preview allows users to review and download their detailed analysis reports.







Results + Future Scope

Results Achieved:

- *Complete working system built and tested*
- *ATS scoring algorithm implemented*
- *Dynamic PDF report generation completed*
- *Professional, responsive UI developed*

Future Scope:

- *LLM Integration for smarter analysis*
- *Job Recommendation System*
- *Resume Ranking capabilities*
- *LinkedIn Profile Analysis*
- *Multi-language Support*

Conclusion

ResumeIQ AI automates resume analysis using:
• Role-Based Evaluation for targeted job matching
• ATS Scoring for recruiter-ready formatting
• Smart Feedback for actionable improvements
• Dynamic PDF Reports for professional documentation
This system empowers candidates to optimize their resumes while supporting modern recruitment workflows.

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Thank You!